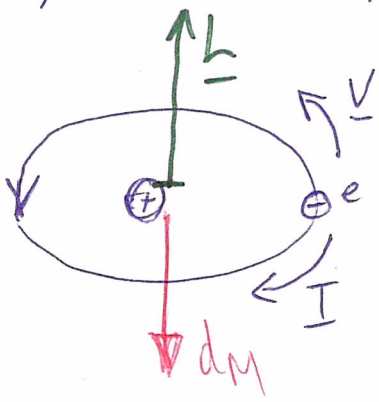
(9.) Magnetism in materials

Simple model of atomic currents:



Semi-classical model of a H atom: e^- orbits around nucleus.

↙ Angular momentum ↘ Magnetic dipole moment

$$\underline{L} = m_e v r \underline{u}_N$$

$$\underline{d}_M = I A(-\underline{u}_n) =$$

$$= \frac{-qv}{2\pi r} \cdot r^{\cancel{2}}\pi \cdot -\underline{u}_n =$$

$$I = \frac{|q \cdot v|}{2\pi r}$$

$$= \frac{qvr}{2} \underline{u}_n \left(= \frac{q m v r}{2m} \cdot \underline{u}_N = \frac{q}{2m} \underline{L} \right) (*)$$

↑ giromagnetic constant.

- Atoms & molecules have so called orbital magnetic moment due to bounded e^-s orbiting.
- In addition, e^-s also have an own magnetic moment called spin magnetic moment (and spin angular moment)

Note: e^- is not spinning! point like

(*) Bohr model: L is quantized, integer multiple of $\hbar$

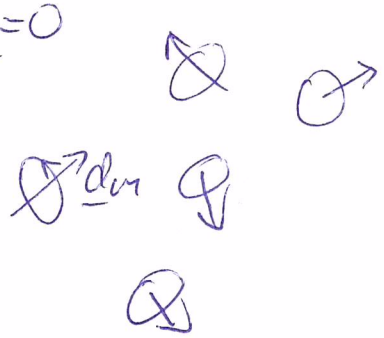
$$\hookrightarrow |\underline{d}_m| = \left| \frac{q}{2m} \underline{L} \right| = \cancel{R} \left(\frac{q\hbar}{2m} \right)$$

↑ Bohr magneton (smallest magn. moment)

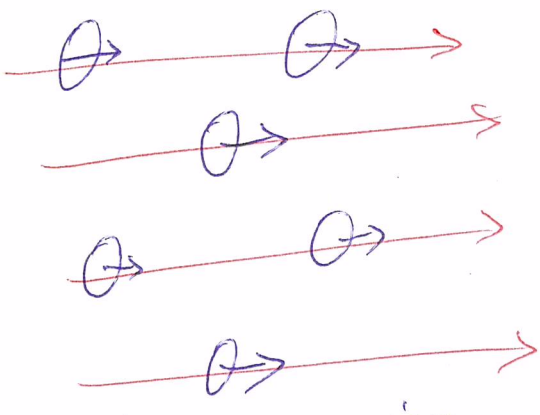
(formula 29.21 $\mu = \cancel{\ell} \frac{e}{2m} \hbar$ ↑ $= 9.27 \cdot 10^{-24}\ Am^2$)

Our model:

$\underline{B}=0$



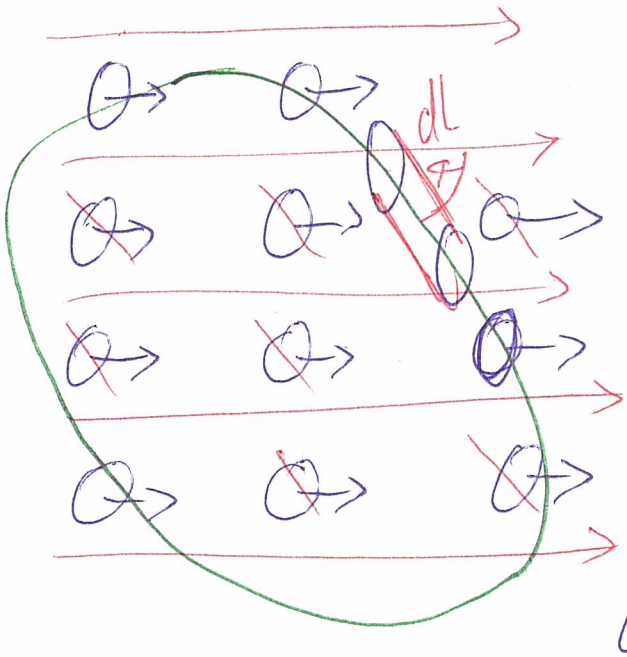
$\underline{B}$



Average magnetic moments growing linearly with B (counterforces: thermal movement & dipol-dipol int.)

Gauss's law: $\oint \underline{B}\,d\underline{s} = 0$ ↖ OK

Ampère's law: $\oint \underline{B}\,d\underline{\ell} = \mu_0 I = \mu_0 (I_{free} + I_{bound})$



$I_{bound} = ?$

$dN_{MD} = n \cdot dV = n\,dl \cdot A_{MD} \cos\alpha$

$dI_{bound} = \underbrace{I_{MD} A_{MD}}_{|\underline{d}_{MD}|} \cdot n\,\underbrace{dl \cdot \cos\alpha}_{|d\underline{\ell}|}$

$dI_b = n \cdot \underline{d}_{MD} \cdot d\underline{\ell}$

$\underbrace{n \cdot \underline{d}_{MD}}_{:=\underline{M}}$ Magnetic dipole moment density (magnetization)

$I_b = \oint \underline{M}\,d\underline{\ell}$ ↙

So Ampère's law:

$$\oint \underline{B}\, d\underline{r} = \mu_0 I_f + \mu_0 \oint \underline{M}\, d\underline{r}$$

$$\oint \left(\underline{B} - \mu_0 \underline{M}\right) d\underline{r} = \mu_0 I_f$$

$$\oint \underbrace{\left(\frac{\underline{B}}{\mu_0} - \underline{M}\right)}_{\underline{H}} d\underline{r} = I_f$$

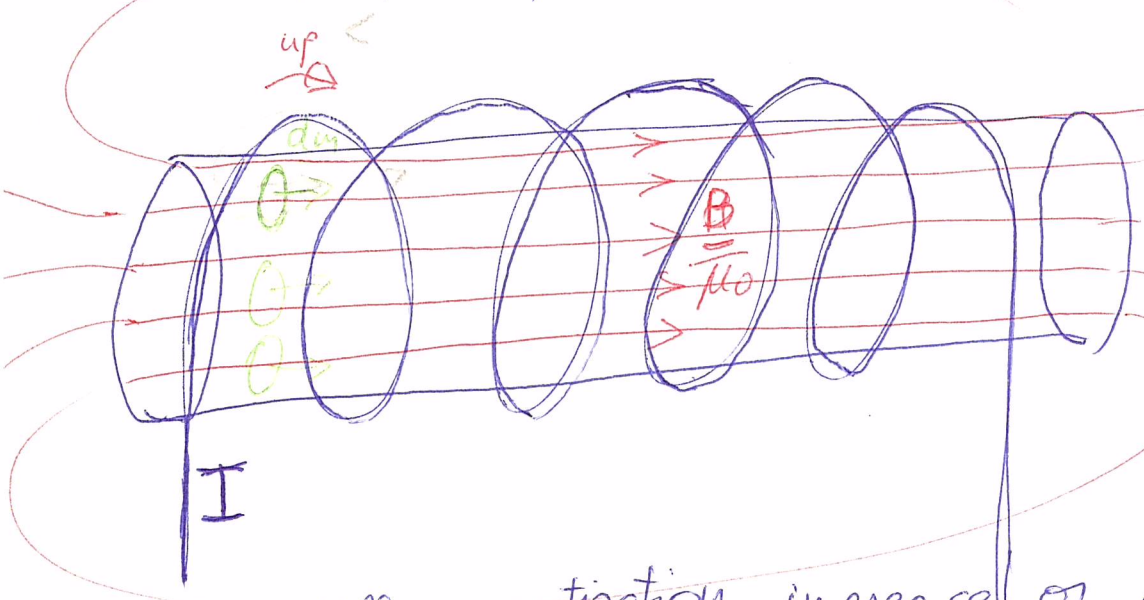
$\underline{H}$: Magnetic field strength generated by free currents

$$\underline{H} = \frac{\underline{B}}{\mu_0} - \underline{M} \qquad [H] = \frac{A}{m}$$

Ampère's law $\oint \underline{H}\, d\underline{r} = I \overset{\text{Stokes}}{\Longleftrightarrow} \operatorname{curl} \underline{H} = \underline{j}$

Free currents generate curly (rotational) magn. field.

$$\underline{B} = \mu_0 (\underline{H} + \underline{M})$$



In a solenoid H is the same regardless of material.

B increases in the material (for most mat's)

Will magnetization increase or decrease magnetic induction? Increases!

Behaviour of materials in magn. field

Most materials follow the model:

$$\underline{B} = \mu_0 (H + M)$$

(Convention: H is considered as independent variable, as it depends only on free currents)

Hom. isotropic lin. mat:

$$\underline{M} = \chi_M \underline{H}$$

χ_M ← magnetic susceptibility

$$\underline{B} = \mu_0 H + \mu_0 \chi_M H = \mu_0 (1 + \chi_M) H = \mu_0 \mu_{rel} H = \mu H$$

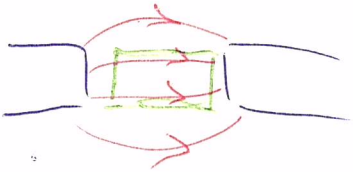
μ ← permeability

(1) For most materials $\chi_M > 0$ $(10^{-3} - 10^{-6})$ small magnetization ∥ Field, increasing B these are called paramagnetic materials

these materials like to move towards strong magn. field,

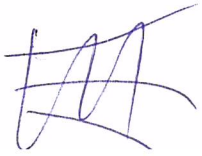


O_2 ↓ paramagnetic!

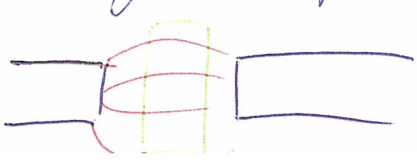


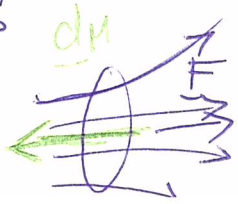
N_2, H_2: Dia !

(2) Surprise:
Some materials (Bi, Mg, Cu, H_2O, N_2, H_2) show negative χ_M (-10^{-6}), small effect, but decreases the induction. These materials escape from magnetic field!



$d_M \uparrow\downarrow H, B$





These are called diamagnetic materials.

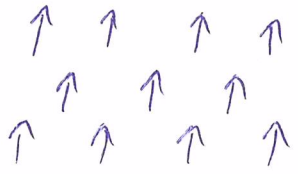
(3) Surprise No 2: Co, Gd

Some materials (Fe, Ni, Nd, Cr, Sm) show very strong nonlinear magnetic properties, that can remain without external field! Ferromagnets

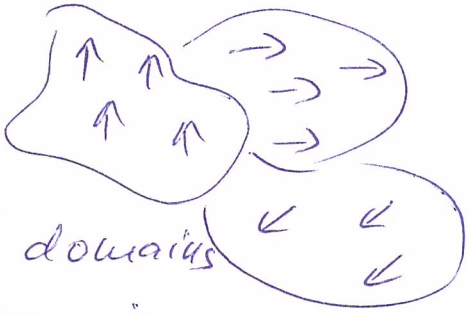
Explanation:

There is a "magical" strong but very short range interaction between the spin magnetic moments of neighbouring Fe atoms, that aligns them // to each other. (The magic interaction is called exchange interaction, can not be explained without quantum mechanics. It relates the unpaired e⁻ spins of the d orbits)

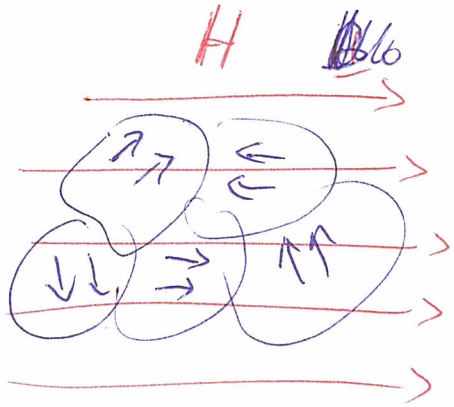
Short range order



Long range disorder (dipole-dipole int.) thermal movement



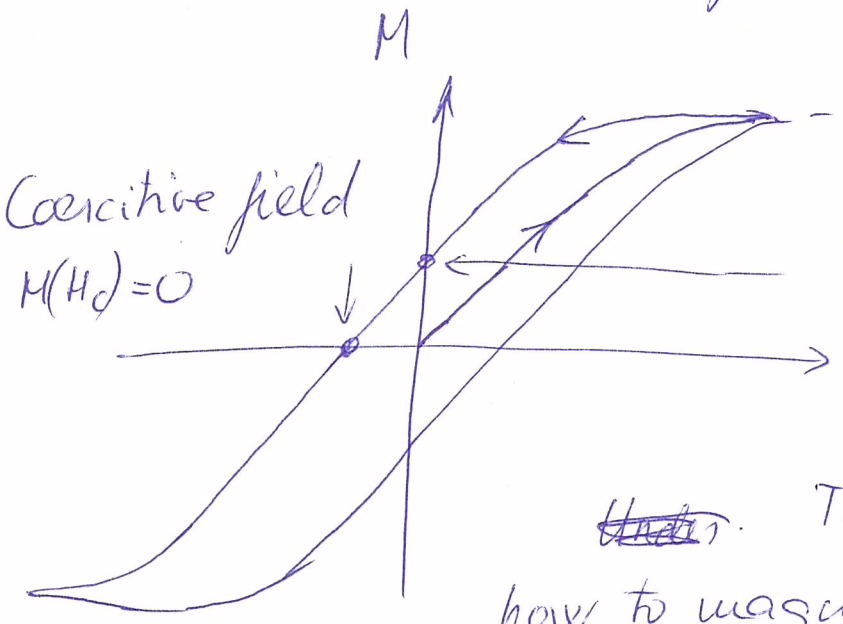
Domains of aligned magnetic moments: Magnetic domains or Weiss domains



In external magnetic field: H

① Whole domains switch to ∥ H

② ∥ aligned domains start to grow (to the detriment of others)



↖ Saturation of magnetization

Remanent magnetization

Hysteresis loop

This model explains how to magnetize ferromagnets.

But how to de-magnetize them?

① Hysteresis looping with decreasing H (Magn tapes)

② Heating above Curie temperature (Hard disk ?)

(Fe: 1043 K, Ni 627 K)

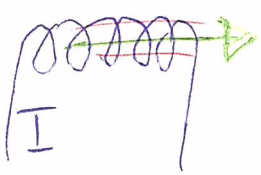
- Soft iron: narrow hysteresis, requires small energy to switch (iron core in el. magnets)
- Hard magnet: wide hysteresis, huge energy is needed to switch → Permanent magnets

Large effect: $\left(\frac{dM}{dH}\right)_{H=0}$ can be $10^4 - 10^5$ (99,95% annealed) (Fe: 200 000)

↑

Induction heaters

In vacuum



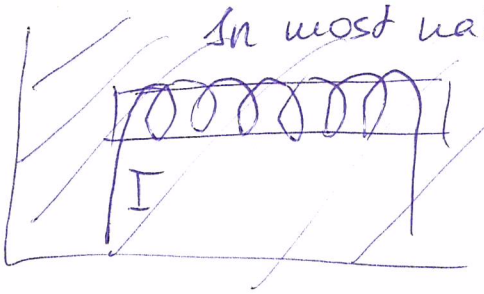
$\oint H\,dl = I$

$H \cdot l = N \cdot I$

$$H = \frac{NI}{l}$$

$\underline{B}_0 = \mu_0 \underline{H}$

In most materials



homogenous

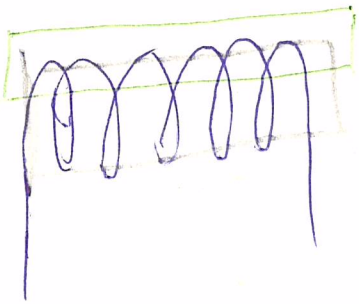
H is the same:

$$H = \frac{NI}{l}$$

but

$\underline{B}$ ~~inside~~ $= \mu_0 \underline{H} + \mu_0 \underline{M}$

If only a part is filled with material:



Boundary conditions !

$\oint \underline{H}\,d\underline{r} = \underline{I} \rightarrow H_{\parallel 1} = H_{\parallel 2}$

$\oint \underline{B}\,d\underline{s} = 0 \rightarrow B_{N1} = B_{N2}$

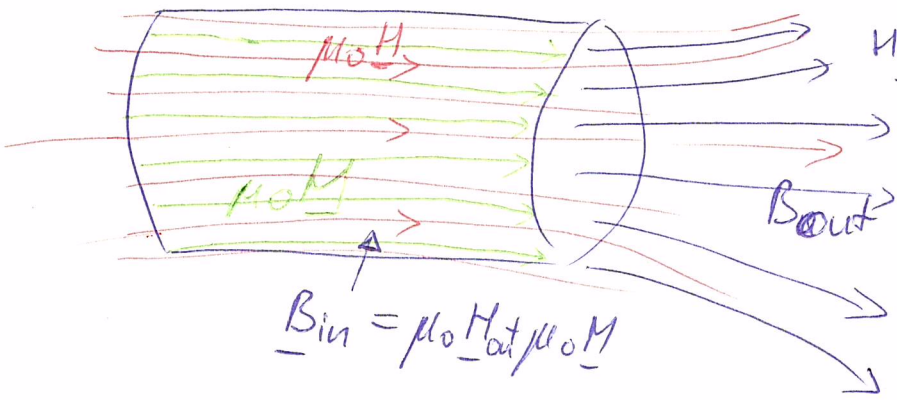
B grows inside → it also grows outside !

outside $\underline{H} = \frac{\underline{B}}{\mu_0}$ H also grows outside !

but $\oint \underline{H}\,d\underline{r} = \underline{I} = \text{Konst}$: H decreases inside !

(↔ scattered field more imp.

← B : continous



H : increases ! outside !

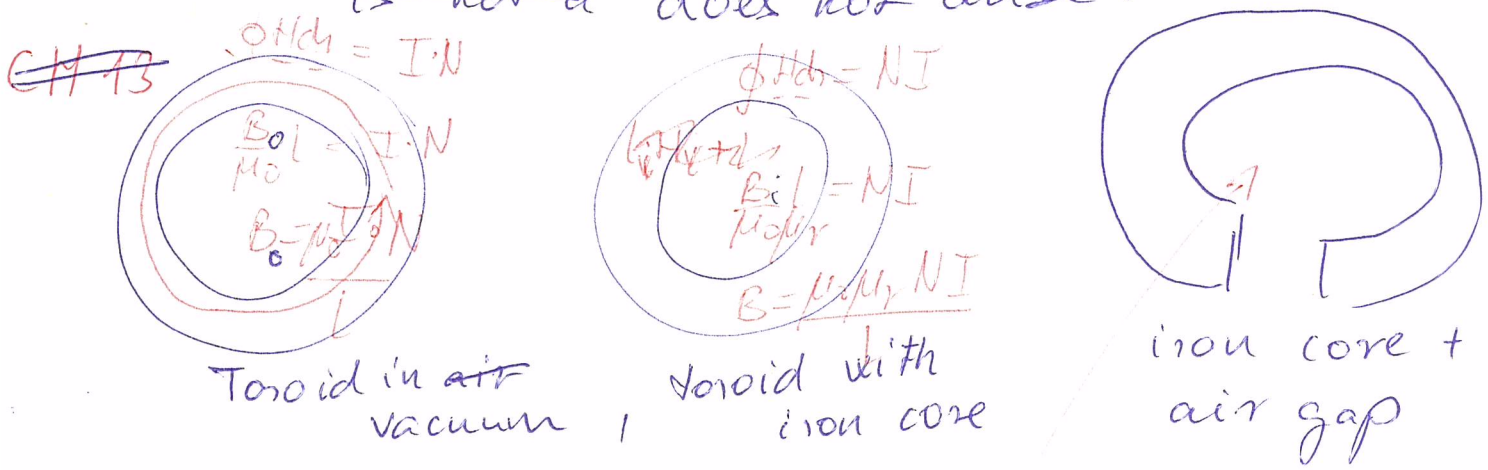
decreases inside

$B_{out} = \mu_0 H_{out}$

Practical difficulty:

We have neglected the scattered field!

For an infinitely long solenoid the problem ~~is not a~~ does not arise.



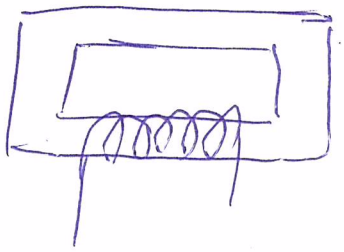
Toroid in ~~air~~ vacuum

Toroid with iron core

iron core + air gap

Conclusion: Iron core is effective if the whole volume (where magn. field is) filled with iron.

Cheaper alternative: closed magnetic loop.



Grand majority of induction lines will remain in the iron!

$$\oint H dl = NI$$

$$H_i l_i + H_{ai} l_a = NI$$

$$\frac{B}{\mu_r \mu_0} l_i + \frac{B}{\mu_0} l_a = NI$$

$$B = \frac{\mu_0 NI}{\frac{l_i}{\mu_r} + l_a}$$

we know:

$$B_{i\perp} = B_{a\perp}$$

$$B_i = B_a \quad (H_i \neq H_a)$$

E.g. $l_i = l_a = \frac{l}{2}$

$$B = \frac{\mu_0 \mu_r NI}{l/2(1+\mu_r)} \approx \frac{2\mu_0 NI}{l}$$